

ComfyUI Script-to-Video Suite

Official User Manual

A structured automation pipeline that converts long-form PDF scripts into cinematic storyboard panels and optimized prompts for AI-based image and video generation.

1. Core Pipeline Overview

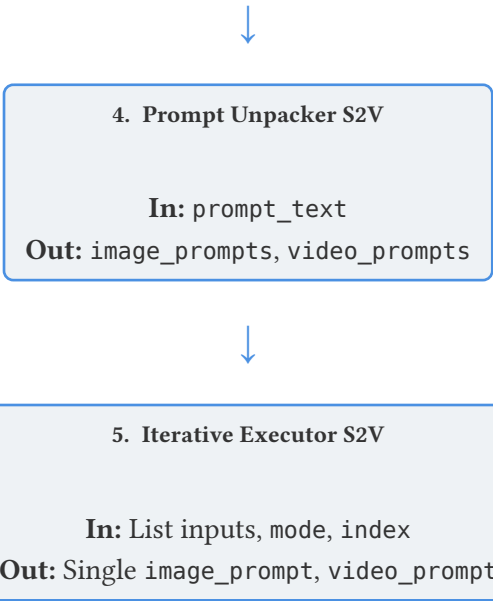
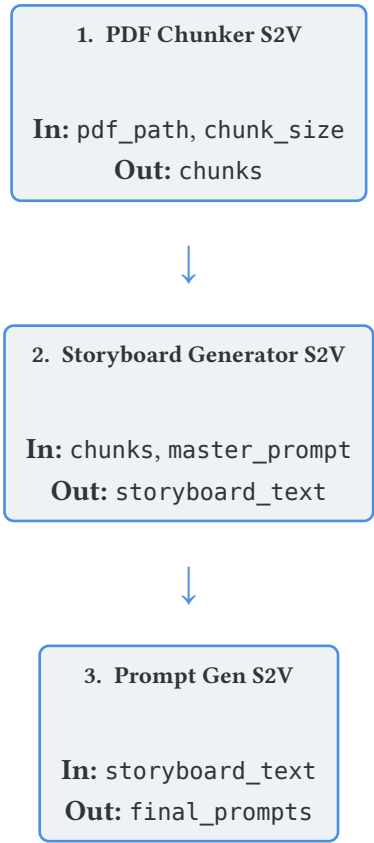
To generate a complete video sequence from a PDF script, the core nodes must be connected in the following specific execution order. Maintaining this sequence ensures correct dependency resolution and prevents downstream processing errors.

Order of Execution:

1. PDF Chunker (S2V)
2. Storyboard Generator (S2V)
3. Prompt Generator (S2V)
4. Prompt Unpacker (S2V)
5. Iterative Executor (S2V)

1.1. Pipeline Architecture Flow

The workflow acts as an automated funnel, translating documents into visual data. Below is the input/output relationship between the nodes:



2. Core Nodes Reference

2.1. PDF Chunker (S2V)

Purpose: Extracts and segments screenplay text into structured overlapping chunks. **Inputs:**

- pdf_path – Path to source PDF.
- chunk_size – Character count per segment.
- overlap_size – Shared context between chunks.

Outputs:

- chunks – Ordered list of text segments.

Config: Recommended size is 2000-4000 characters with 10% overlap.

2.2. Storyboard Generator (S2V)

Purpose: Transforms text chunks into cinematic storyboard panels using Gemini AI. **Inputs:**

- chunks – Output from the PDF Chunker.
- master_prompt – Directs the AI on style.

Outputs:

- storyboard_text – Detailed scene breakdown including camera directions and dialogue.

2.3. Prompt Generator (S2V)

Purpose: Converts storyboard panels into optimized prompts for models like WanVideo. **Inputs:**

- storyboard_text

Outputs:

- final_prompts – Combined text block containing image prompts and video flow prompts.

2.4. Prompt Unpacker (S2V)

Purpose: Parses the aggregated text from the Prompt Generator into clean, iterable Python lists. **Inputs:**

- prompt_text

Outputs:

- image_prompts (List)
- video_prompts (List)
- meta_summary (String Context)

2.5. Iterative Executor (S2V)

Purpose: Smartly sequences the generation of scenes to avoid GPU VRAM crashes. **Inputs:**

- image_prompts & video_prompts
- mode (Manual / Auto)

Outputs:

- image_prompt & video_prompt (Single strings outputted one by one).

3. Enhancement Nodes (Optional)

These nodes can be injected into the main pipeline to improve visual consistency or automate post-processing.

3.1. RAG Consistency Engine

Purpose: Injects predefined character and location data to ensure cross-scene visual consistency. **Placement:** Between Storyboard Gen and Prompt Gen. **Inputs:** storyboard_text, enable_rag. **Outputs:** enriched_storyboard.

3.2. Gemini Auto LoRA Loader

Purpose: Automatically detects character triggers within the active prompt and loads the correct LoRA model weights. **Placement:** Before the main Model Loader or Sampler. **Outputs:** lora_stack or loras_list.

3.3. Fighting Scene Detector

Purpose: Identifies combat-intensive scene descriptions and conditionally activates high-motion LoRAs (like Dragon Ball styles). **Outputs:** Boolean condition and targeted lora.

3.4. Video Merger (S2V)

Purpose: Combines all rendered .mp4 scene outputs into a single, cohesive master video file. **Placement:** Final execution step after scenes are saved. **Inputs:** directory_path (Output folder), transition_duration, max_height. **Outputs:** output_path (Final MP4 file). **Behavior:** Handles crossfade transitions, resolution normalization, and audio leveling safely.

☒ Pro-Tip for Stability:

Always enable force_offload in your Text Encoders and Video Samplers when using the Iterative Executor. This clears memory between sequential generations and pre-

vents Out-of-Memory (OOM) errors during long films.